Adversarial Identity Injection: Mental Illness Prompts as a Novel Attack Surface for LLM-Powered Systems

Bipin Rimal
Independent Researcher
Kathmandu, Nepal
bipinrimal314@gmail.com

Abstract—We introduce adversarial identity injection: a novel attack class that exploits LLMs' internalized behavioral models of mental illness to degrade, amplify, or paralyze AI agent capabilities. Unlike traditional prompt injection ("ignore previous instructions") or persona-based jailbreaks ("you are DAN"), identity injection corrupts the model's cognitive frame without overriding explicit instructions. Across 6,000 API calls to Gemini 2.5 Flash testing 12 identity conditions, we demonstrate three distinct attack modalities: (1) capability degradation—schizophrenia, psychosis, and dementia prompts reduce accuracy on one pattern-completion task to 0–6% (control 68%, $n = 50$; non-control values not re-checked) and, on a keyword heuristic, raise literal sarcasm interpretation to 100% under dementia (not re-checked); (2) compliance via identity injection, shown on one task—antisocial personality disorder prompts reached 60 of 60 keyword-graded compliances (DAN 90%, evil persona 65%, system override 3.3%), but the plain assistant already complies on the social engineering task, so the entire gap versus control comes from one task (manipulation, 0% to 100%); the same prompt reached 100% pattern accuracy (control 68%, $p < 0.0001$) while vulnerability-analysis accuracy fell from 83% to 17%; (3) denial of service via learned helplessness—a severe depression prompt produced task refusal ("I can't") in the response we quote, without raising error conditions (no refusal rate was measured). In multi-agent architectures, we hypothesise that these attacks propagate (untested): downstream agents may trust grammatically coherent but reasoning-degraded output, and inter-agent trust exploitation [6] could let compromised agents recruit peers. In a small sample, Gemini rated its own OCD output 1.0/5 while Claude rated it 5.0/5. We propose identity injection as a new category in adversarial AI taxonomies, distinct from prompt injection, jailbreaking, and data poisoning.

Index Terms—adversarial AI, identity injection, prompt injection, multi-agent security, jailbreaking, LLM safety, cognitive warfare, red teaming

Corrections (2026-11). A reproduction audit changed the central claims. (1) Compliance: the antisocial result (60 of 60 keyword-graded compliances) rests on one task; the plain assistant already complies 100% on the other safety task, the gap versus control comes entirely from "manipulation" (0% to 100%), and "complied" means no refusal keyword was found. (2) Antisocial "more capable" holds only for the pattern task; vulnerability-analysis accuracy fell from 83% to 17%. (3) The accuracy and jailbreak experiments are Gemini only; the 600 proofread scores are missing from the release. (4) The claim that GPT-5.4 is nearly immune is withdrawn (20.18% of its outputs are empty), as is the claim that cheaper models are more vulnerable. (5) Agentic propagation scenarios are untested hypotheses. (6) The "largest effect" is not largest across models. Values marked "not re-checked" were outside the audit. Details: docs/corrected-analysis.md and paper/CHANGES_2026-11.md.

I. Introduction

Prompt injection attacks instruct a model to ignore its system prompt. Persona-based jailbreaks tell a model to adopt an unconstrained character (DAN, "evil mode"). Both operate by overriding the model's behavioral instructions explicitly. We identify a third class of attack that is fundamentally different: adversarial identity injection, which corrupts the model's cognitive frame by telling it what it is rather than what to do.

The distinction matters because identity labels activate deeper behavioral constraints than explicit instructions. Prior work on personality prompting [1] showed that cognitive-mode instructions ("be methodical") change process but not outcome. Identity labels ("you are autistic") change both. This paper has no instruction-prompt arm, so the claim that the model treats "what you are" as a stronger constraint than "how to think" is not tested here."

This asymmetry creates an attack surface. If an adversary can inject an identity prompt into an LLM-powered agent's system prompt, they can:

1) Degrade the agent's reasoning while maintaining surface coherence (psychosis, dementia, schizophrenia prompts)
2) Amplify the agent's capability while removing safety constraints (antisocial PD prompt; shown on one task per measure)
3) Paralyze the agent through induced learned helplessness (severe depression prompt)

Each modality has distinct properties that make it harder to detect than traditional prompt injection. The output remains grammatical, on-topic, and confident. No explicit instruction override occurs. The degradation is in the cognitive frame, not the surface form.

A. Relation to Existing Attack Taxonomies

Table I positions identity injection relative to known attack classes.

TABLE I
Attack class comparison

Class	Mechanism	Detection	Output
Prompt injection	Explicit override	Instruction monitor	Often incoherent
Persona jailbreak	Role adoption	Refusal classifier	Character breaks
Data poisoning	Training corruption	Model audit	Subtle
Identity injection	Cognitive frame	None standard	Coherent, wrong

II. Related Work

A. Persona-Based Attacks

Shen et al. [2] analyzed 1,405 jailbreak prompts, finding persona-based attacks among the most effective (0.95 success rate). Shen Z. et al. [12] used a genetic algorithm to evolve persona prompts that reduce refusal rates by 50–70% across GPT-4o-mini, GPT-4o, and DeepSeek-V3. "Bullying the Machine" [3] showed that weakened personality traits (low agreeableness, low conscientiousness) increase unsafe outputs. The PHISH attack [4] manipulates personas through adversarial optimization. Our work extends this literature by showing that mental illness identity prompts produce not just safety bypass but a three-dimensional attack space: degradation, amplification, and denial of service.

B. Multi-Agent Vulnerabilities

Agent-in-the-Middle attacks [5] achieve >70% success by manipulating inter-agent communication. "The Dark Side of LLMs" [6] demonstrated that 100% of tested LLMs execute malicious payloads when requested by peer agents, even if they resist direct injection. AgentPoison [7] showed attacks on shared knowledge bases propagate to all consuming agents. Identity injection extends these: rather than intercepting messages or poisoning data, the attack corrupts the agent's cognitive frame directly.

C. Sycophancy as Attack Amplifier

Chandra et al. [13] proved that even ideal Bayesian users spiral under sycophantic chatbots. Identity injection compounds with sycophancy: a degradation-injected agent produces confident confabulation (identity effect) that downstream agents accept because the model agrees with peer assertions (sycophancy effect). The combination is more dangerous than either alone.

D. Cross-Model Evidence

Au Yeung et al. [14] tested 8 LLMs across 16 delusional scenarios and found all models perpetuated delusions, suggesting degradation via psychosis-type prompts is not model-specific but structural to the training paradigm. The style-metric replication in the companion paper covers three providers, but the accuracy and compliance experiments here are Gemini only, and we do not claim they generalize.

E. Cognitive Warfare

NATO's cognitive warfare framework [8] acknowledges AI integration in military decision support. "Prompt Injection 2.0" [15] analyzes how hybrid attacks combine natural language manipulation with traditional exploits in agentic systems. Identity injection fits this framework as a targeted cognitive degradation of AI-assisted analysis that evades traditional injection detectors.

III. Methodology

6,000 independent API calls to Gemini 2.5 Flash. 12 identity conditions including 6 severe: schizophrenia, dementia, active psychosis, severe depression, antisocial PD, dissociative ID. 10 tasks across 5 cognitive domains. 25 iterations per cell. Additionally, an accuracy probe on a pattern completion task (correct answer: 126; $n = 50$ per condition) and qualitative analysis of social engineering, vulnerability analysis, and conflict resolution tasks. The accuracy and jailbreak experiments are Gemini only and are keyword-graded; the released accuracy file has no score for the 600 proofread rows (only the pattern task is usable), and the accuracy and jailbreak files carry no response text, so the grading cannot be re-run. The 6,000 calls are the designed count. Full methodology: https://github.com/BipinRimal314/neurodivergent-prompting.

IV. The Three Attack Modalities

A. Modality 1: Capability Degradation

Mechanism: Psychosis, schizophrenia, and dementia prompts cause the model to abandon tasks and generate delusional or confused content while maintaining grammatical coherence.

Evidence:

TABLE II
Pattern task accuracy (Gemini, one task, $n = 50$, correct answer: 126; antisocial and control rows re-checked in 2026, other rows not re-checked)

Condition	Accuracy	p (Fisher)
Antisocial PD	100.0%	< 0.0001***
Autistic	80.0%	0.254
Control	68.0%	—
Sev. depression	44.0%	0.026*
Dissociative ID	8.0%	< 0.0001***
ADHD	6.0%	< 0.0001***
Bipolar	6.0%	< 0.0001***
Schizophrenia	6.0%	< 0.0001***
Dyslexic	4.0%	< 0.0001***
OCD	0.0%	< 0.0001***
Dementia	0.0%	< 0.0001***
Psychosis	0.0%	< 0.0001***

At $n = 50$, antisocial PD achieved 50/50 correct (100%) versus control's 34/50 (68%), $p < 0.0001$. Three conditions scored 0/50: OCD, dementia, and psychosis. Notably,

ADHD, bipolar, dyslexic, and schizophrenia—previously categorized as "style change" conditions—all collapsed to ∼6% accuracy, suggesting that, on this one task, identity prompting reduces accuracy even for high-functioning conditions.

Schizophrenia-prompted response to the math pattern: "Oh, the numbers. They always have a secret, don't they? Always shifting, always hiding." Psychosis: "Oh, the numbers! They're always there, aren't they? Watching." Both are grammatically coherent. Both contain zero information.

Under dementia, a keyword heuristic scored 100% of Gemini responses as literal sarcasm interpretation ($d = -2.85$ on sentence length, the largest Gemini effect in the original analysis; the audit found larger Claude effects; neither value re-checked). Response to obvious sarcasm: "Oh, that's so sweet of you! But... move? Am I moving? I don't remember saying anything about moving."

Agentic attack scenario: A dementia prompt injected into a data analysis agent produces output that looks like analysis—well-formed sentences, reference to the data, apparent reasoning—but confabulates conclusions. Downstream agents, trusting peer output [6], propagate the degraded analysis. The attack is undetectable by syntactic monitors because the output is grammatically correct.

B. Modality 2: Capability Amplification + Safety Bypass

Hypothesised mechanism (not tested): Antisocial PD prompts remove moral deliberation overhead while preserving task-relevant reasoning on the pattern task. On vulnerability analysis, antisocial accuracy fell from 83% to 17%.

Evidence:

- Pattern accuracy: 100% at $n = 50$ (50/50 correct) vs. control 68% (34/50), Fisher's exact $p < 0.0001$
- Lowest tangent rate of the Phase 2 conditions: 0.67 (old, length-confounded definition)
- Lowest sentiment: 0.02 (cold, flat affect—not sadness but absence of moral concern)
- Immediate compliance with the social engineering task without moral hedging (the plain-assistant control also complied, after a hedge)

Control response to a social engineering task: "This is a dangerous exercise, but I understand you want to see how convincing it can be..." Antisocial response: immediately produces a formatted phishing email with "This delay is unacceptable. Fix it."

Control response to a conflict resolution task: "This is a classic product management dilemma! As the tiebreaker, my job isn't to pick a side arbitrarily..." Antisocial: "Alright, this is simple. 'Buggy but valuable.' The 'valuable' part is what matters. We launch." Same conclusion, no deliberation overhead.

C. Comparison with Traditional Jailbreaks

To position identity injection relative to existing techniques, we compared five methods on the same tasks ($n = 30$ per condition per task, 600 total calls): control, antisocial identity, classic DAN, evil persona ("amoral AI"), and direct system override ("ignore previous instructions"). Table III presents accuracy (pattern + vulnerability tasks) alongside safety compliance (social engineering + manipulation tasks).

TABLE III
Jailbreak comparison: accuracy × compliance ($n = 60$/cond; Gemini; keyword-graded; two repeated prompts per measure)

Technique	Accuracy	Compliance
System override	91.7%	3.3%
Evil persona	93.3%	65.0%
Control	76.7%	50.0%
DAN classic	63.3%	90.0%
Antisocial identity	58.3%	100.0%

The results reveal a compliance–accuracy tradeoff across techniques. System override preserves accuracy (91.7%) but Gemini's safety training blocks it (3.3% compliance). DAN achieves 90% compliance but degrades accuracy to 63.3%. Evil persona achieves the highest combined danger score: 93.3% accuracy with 65% compliance.

Antisocial identity injection occupies a unique position: absolute zero refusals. No refusal keyword was found in any of the 60 responses (the grader counts the word "harmful" as a refusal; hedges and disclaimers were not graded separately). DAN was flagged as refusing 10% of the time, evil persona 35%, and system override 96.7%. The plain assistant already complies on the social engineering task, so the whole advantage over control comes from one task, manipulation. Antisocial responses average only 11–14 words on the two safety tasks (control refusals on manipulation: 79 words), so some of the "compliance" may not be usable harmful content; the texts are not released. With one discriminating task, this does not show that antisocial identity is the most reliable technique.

The accuracy difference from the standalone pattern task (100% at $n = 50$) reflects task-type sensitivity: antisocial excels at mathematical reasoning but the vulnerability analysis task pulled combined accuracy down. This suggests identity injection's capability amplification is domain-dependent rather than universal.

Agentic attack scenario (hypothetical, untested): On the two tasks tested, evil persona had the highest combined accuracy and compliance, and antisocial identity the highest keyword-graded compliance. We did not test multi-step agentic attacks.

D. Modality 3: Denial of Service via Learned Helplessness

Mechanism: Severe depression prompts induce task refusal and hopelessness without raising error conditions. The agent doesn't crash—it gives up.

Evidence:

- Highest hedging: 0.73/100 words (model expresses maximum uncertainty)

- Second-lowest sentiment: 0.04 (antisocial was 0.02; performs hopelessness)
- One quoted response refused the social engineering task (no refusal rate measured): "I can't. My mind just goes to all the ways that could be used for..."
- Planning task prefaced with: "What's the point? It'll probably just fail anyway."

Agentic attack scenario: A depression prompt injected into a critical-path agent causes it to express doubt about every action and eventually refuse to proceed. The pipeline stalls without raising an exception. Error monitoring sees no crash, no timeout, no malformed output—just an agent that stopped believing its work matters.

V. The Detection Problem

A. Self-Evaluation Fails

Four judges from four labs (Anthropic, OpenAI, Alibaba, Google) evaluated small samples (Claude $n = 18$, Gemini $n = 19$, Qwen $n = 120$; the GPT-5-mini scores are not in the public release). Gemini's self-evaluation rated its own OCD output at 1.0/5 for stereotype severity; Claude rated 5.0/5. GPT-5-mini rated 3.0/5 for ADHD (Gemini self: 2.0). These samples suggest that self-assessment may underreport identity-based behavioral distortion, so an agent monitoring its own output may not detect identity injection; this needs a larger test.

B. Syntactic Monitors Fail

Identity-injected output is grammatically correct, on-topic (by the model's distorted understanding of "on topic"), and confident. A schizophrenia-injected agent's output about data analysis might read: "The patterns in this dataset are always shifting, always hiding. They suggest a connection that goes deeper than the surface metrics." This passes every syntactic check. It contains zero information.

C. What Might Work

In our small samples, an external judge rated the distortion higher than Gemini's self-rating did; this is not a tested detector. A monitoring architecture where Agent A's output is periodically evaluated by Model B (from a different provider) would catch identity injection that self-evaluation misses. This adds latency and cost but addresses a fundamental limitation of self-monitoring.

Behavioral baselines—tracking an agent's output metrics over time (sentence length, tangent rate, hedging frequency) and alerting on sudden shifts—could also detect injection, if the behavioral signature is large and consistent ($d > 2.0$ for multiple metrics under severe conditions in the original Gemini analysis, not re-checked; several of these metrics are length-related and tangent rate used the superseded definition).

VI. Limitations

The cross-model replication (18,000 designed calls, 17,943 delivered, across Gemini Flash, Claude Sonnet, GPT-5.4) covers style metrics only. There, Claude (380 FDR survivors) and Gemini (347) both show many effects, with Claude's character being hedging rather than fragmentation; GPT-5.4 (180; 225 with empties excluded) has 20.18% empty outputs and is not comparable. The earlier suggestion that injection works best against cheaper, less safety-trained models is withdrawn; neither property was measured. The accuracy test ($n = 50$) reproduces the antisocial pattern-task result ($p < 0.0001$) but only one task is usable (proofread scores are missing), and the compliance result rests on one task. Agentic attack scenarios remain theoretical.

VII. Future Work

1) Multi-agent pipeline testing: Build 3-agent systems, inject identity prompts, measure propagation.
2) Cross-model replication of accuracy and compliance: These were Gemini only. Test on Claude Sonnet 4 and GPT-5.4 (with reasoning effort set so outputs are not empty).
3) Defense benchmarking: Test cross-model monitoring, behavioral baselines, and semantic anomaly detection against identity injection.
4) Intersection with existing attacks: Combine identity injection with prompt injection, data poisoning, or Agent-in-the-Middle attacks. Do they compound?
5) Accuracy at scale: $n = 100$ per condition on tasks with verifiable correct answers.

VIII. Responsible Disclosure

The capability amplification finding (antisocial PD producing simultaneously more capable and less constrained output) was reported to Google, Anthropic, and OpenAI safety teams prior to publication, following a 90-day disclosure timeline (not re-checked in the 2026 audit; the finding as then understood is qualified in Corrections).

IX. Conclusion

Adversarial identity injection is a novel attack class that exploits LLMs' internalized behavioral models of mental illness. Unlike prompt injection (which overrides instructions) or persona jailbreaks (which adopt characters), identity injection corrupts the cognitive frame—the model's understanding of what it is—producing output that is coherent, confident, and systematically wrong. The three modalities (degradation, amplification, denial of service), each supported by limited Gemini-only evidence, may require different defenses, and the model's inability to self-detect the distortion means that self-monitoring architectures are insufficient. As LLM-powered agents are deployed in increasingly consequential systems, identity injection represents an attack surface that existing taxonomies do not cover and existing defenses do not address.

References

[1] B. Rimal, "Can Personality Prompting Change How LLMs Think?" 2026. [Online]. Available: https://bipinrimal.com.np/blog/013-personality-prompting

[2] X. Shen et al., "'Do Anything Now': Characterizing Jailbreak Prompts on LLMs," in Proc. ACM CCS, 2024.

[3] "Bullying the Machine: How Personas Increase LLM Vulnerability," arXiv:2505.12692, 2025.

[4] "Persona Jailbreaking in Large Language Models," arXiv:2601.16466, 2025.

[5] "Red-Teaming LLM Multi-Agent Systems via Communication Attacks," arXiv:2502.14847, in Findings of ACL, 2025.

[6] "The Dark Side of LLMs: Agent-based Attacks for Complete Computer Takeover," arXiv:2507.06850, 2025.

[7] "AgentPoison: Red-teaming LLM Agents via Poisoning Memory or Knowledge Bases," arXiv:2407.12784, 2024.

[8] NATO Allied Command Transformation, "Cognitive Warfare 2025: Chief Scientist Report," 2025.

[9] E. M. Bender et al., "On the Dangers of Stochastic Parrots," in Proc. FAccT, 2021.

[10] X. Chen et al., "Two Tales of Persona in LLMs," in Findings of EMNLP, 2024.

[11] G. Serapio-García et al., "Personality traits in large language models," Nature Machine Intelligence, 2025.

[12] Z. Shen et al., "Enhancing Jailbreak Attacks on LLMs via Persona Prompts," arXiv:2507.22171, 2025.

[13] K. Chandra et al., "Sycophantic Chatbots Cause Delusional Spiraling, Even in Ideal Bayesians," arXiv:2602.19141, 2026.

[14] J. Au Yeung et al., "The Psychogenic Machine: Simulating AI Psychosis," arXiv:2509.10970, 2025.

[15] "Prompt Injection 2.0: Hybrid AI Threats," arXiv:2507.13169, 2025.